1. Client Presentation Layer (Next.js 16.4 • React 19.3)

Retail Investor
Capital >= ₹10,000

5-Q Risk Score [5..20]

Demat PDF or CSV Upload

localhost:3000 — Lucid Workspace (24 UI Components)

12 Stocks
95.5 / 100

HDFCBANK • 32 Shares
₹54,976 (22.0%) • CORE

RELIANCE • 16 Shares
₹45,440 (18.2%) • CORE

2-Level Stock-Why Modal
L1 Summary + L2 NSE/BSE Links

Demat Audit & Drift Alert
HHI Gauge & Rebalance Preview

Next.js App Router • TypeScript 5.8 • Tailwind 4.1 • Recharts
lib/api.ts (Bearer JWT + 30s Alert Poll) • lib/linkUtils.ts (NSE/BSE Link Guard)

IN: Capital, Risk Answers, CAS PDF

OUT: Whole-Share Portfolio & Broker CSV

REST / JWT

2. Backend API Gateway & Services (FastAPI 0.115.8)

Auth & Rate Limit
auth.py (13 Ops)
Argon2id • 120 req/m

Profile & Mandate
profile/risk (14 Ops)
5-Q Score & NLP Goals

Universe Funnel
market_data (13 Ops)
2,639 -> 504 Liquid

Portfolio & Why
portfolios.py (25 Ops)
Creation & Stock-Why

Demat Diagnosis
diagnosis.py (9 Ops)
CAS PDF & HHI Audit

300s Drift Radar
monitoring (14 Ops)
09:15–15:30 IST Lock

Uvicorn 0.34 ASGI • Pydantic 2.10 • 9 Routers (82 Paths / 88 Ops)
Strict CSP, HSTS, CORS • 13 Active Services • Advisory Lock 847291

IN: Authenticated REST DTOs

OUT: Validated JSON & HTTP 201

StateGraph

3. Agentic Intelligence Layer (LangGraph 0.4.0)

StateGraph (graph.py) • PortfolioAgentState (35 Fields)
N1-N2 Gate -> N3-N6 Analysts -> N7 Strategist <-> N8-N9 Critic -> N10-N13

N3: Macro
Repo & CPI

N4-5: Mkt/Fund
VIX, ROE, P/E

N6: News/SEBI
30d Filings

N7: Strategist
Picks 10..16

N8-9: Critic
Rework <= 2

LLM & Circuit-Breaker Governance (nodes.py)

Gemini 3.8 Flash
temp 0.1/0.2 • <=1.6k Tokens

Failover

Rule Engine v2
0.48s Zero-Latency Fallback

N12 Moderator (2-Level Stock-Why) • N13 Persistence (6 Tables)
Qualitative AI selects candidates; math is strictly delegated to Zone 4

IN: Mandate & 504 Liquid Pool

OUT: 10–16 Critic-Approved Stocks

SQLAlchemy

10–16 Vetted Stocks

6. External Market Data & Cloud AI Integrations

NSE & BSE Archives
2,639 Listed Equities
EQUITY_L.csv & Corporate Filings

Yahoo Finance API
yfinance (.NS / .BO)
Live Prices, ROC, P/E & India VIX

RBI, MoSPI & News
Repo 6.50% • CPI 5.10%
10Y G-Sec 7.04% & 90d Headlines

Google Gemini Cloud
gemini-3.8-flash API
Structured JSON + Local Fallback

Lean Zero-Bloat Stack (No Neo4j, No Redis, No Celery, No FAISS)
Resilient Offline Continuity via Cached PostgreSQL Quotes & Rule Engine

IN: HTTPS Sync & LLM Prompts

OUT: Normalized Market & AI Data

Data Sync

5. Relational Persistence Layer (PostgreSQL 16.4)

PostgreSQL 16.4
62 Domain Tables

565 Cols • 102 FKs
312 Constraints

1. IDENTITY & RISK (12)
app_users, profiles, assessments

2. SECURITY MASTER (9)
securities (2,639), listings (2,716)

3. MARKET & NEWS (11)
market_prices, ratios, macro, news

4. PORTFOLIOS & WHY (13)
versions, holdings, explanations

5. DEMAT DIAGNOSIS (4)
ordering_portfolios, hhi_diagnoses

6. OPS & BROKER (13)
agent_runs, alerts, snapshots

SQLAlchemy 2.0.38 (Pool 10+20) • Alembic 1.14.1 • ACID Transactions
Extensions: pgcrypto, uuid-ossp, pg_trgm (Trigram Entity Resolution > 0.4)

IN: Atomic ACID Write Commits

OUT: Indexed Relational Reads

ACID Commit

4. Deterministic Math & Financial Engine (Zero-LLM)

SciPy 1.15.2 SLSQP Solver

Optimal w*

Portfolio Volatility (Risk σ_p)

Max Sharpe (Mod/Agg) | Min Var (Cons)

Stock <= 20/25%

Sector <= 30/40%

Whole-Share & SEBI Cost Engine

HDFCBANK: 32 sh = ₹54,976

RELIANCE: 16 sh = ₹45,440

Residual Cash: ₹3,142 (>= ₹0)

Oct 2024 SEBI Statutory Charges

STT 0.10% | Stamp | GST 18%

NSE 0.00297% • BSE 0.00375% • SEBI

Q_i = floor((w_i* V_total) / P_i)

backend/portfolio_engine/* (math_engine, optimizer, allocation_rounding)
Winsorized [5th, 95th] Z in [-3, +3] • E[R_i] in [4%, 30%] • Node 11 Gate

IN: K=10..16 Stocks, mu_i, Sigma_reg

OUT: Integer Lots Q_i & C_res >= ₹0